“Write here, write now”: Spelling difficulty disrupts parallel sentence planning

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Our understanding of real-time language planning and, in particular, spelling retrieval in writing is limited. Writers may plan upcoming information at linguistic edges or in parallel to production. We tested whether spelling difficulty disrupts parallel planning of upcoming lexical information.

In three experiments participants (Ns$_{\text{ppts}}$ = 56, 80, 96) were asked to describe spatial arrangements of images (N_{items} = 36) with typed utterances in English of the form *The N1 and the N2 are above the N3*; see Fig. 1. We manipulated N1 spelling difficulty (difficult, e.g. *caterpillar* vs. e.g. *watermelon*, determined from independent naming data, N = 100), and length (1 vs. 2–4 syllables), to reduce time for parallel planning (e.g. *the fly*). Prior to response, participants were primed by seeing displays with the same N1 and the same arrangement as the target, and hearing the image description in the required syntactic form. The prime of half of the items was of the form *The N1 is above …*). To further constrain parallel planning, Exp. 2 elicited sentences without N2-initial determiner (*N1 and N2 are …*). In Exp. 3 N2 was also included in the prime to reduce N2 naming demands. Items were counterbalanced across six (Exp. 1), nine (Exp. 2), and twelve (Exp. 3) lists, presented in randomised order intermingled with 24 filler arrays. We recorded time of each keypress and data modelling focused on determining hesitation to fluent output.

Typing hesitations were extracted using Bayesian mixed-effects mixture models (Roeser et al., 2025). Posterior means are shown in Fig. 2. Difficult-long N1s were associated with more hesitations (compared to easy-long N1s; in all experiments) before, during and after typing N1 (all $P(\hat{\theta} < 0)$s < .01). In Exp. 1, no hesitations were observed immediately before N2 ($P(\hat{\theta} < 0)$s = .35, .49). When the N2 determiner was omitted (Exp. 2), hesitations before N2 increased after short and difficult-to-spell N1s (both $P(\hat{\theta} < 0)$s < .05) but less so for long N1s ($P(\hat{\theta} < 0)$ = .19). N2 preview (Exp. 3) reduced hesitations immediately before N2 for short and difficult-to-spell N1s (both $P(\hat{\theta} < 0)$s > .99) but less so for long N1s ($P(\hat{\theta} < 0)$ = .8).

We found that difficulty with spelling retrieval leads to disruptions of the writing flow. Difficulty with spelling affected the retrieval of N2 but only when there was not enough time for parallel planning. These results demonstrate that written sentence planning unfolds in parallel to production; difficulty with low-level properties (orthography) create a planning bottleneck has a knock-on effect on higher level planning (lexical retrieval of N2).

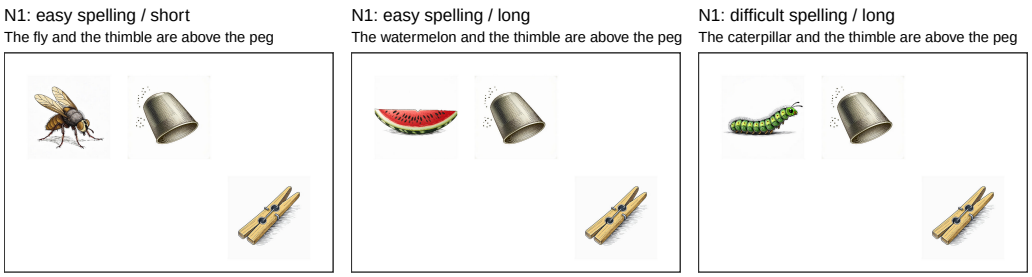

Figure 1: Example array for each N1 type. Target sentences are shown above.

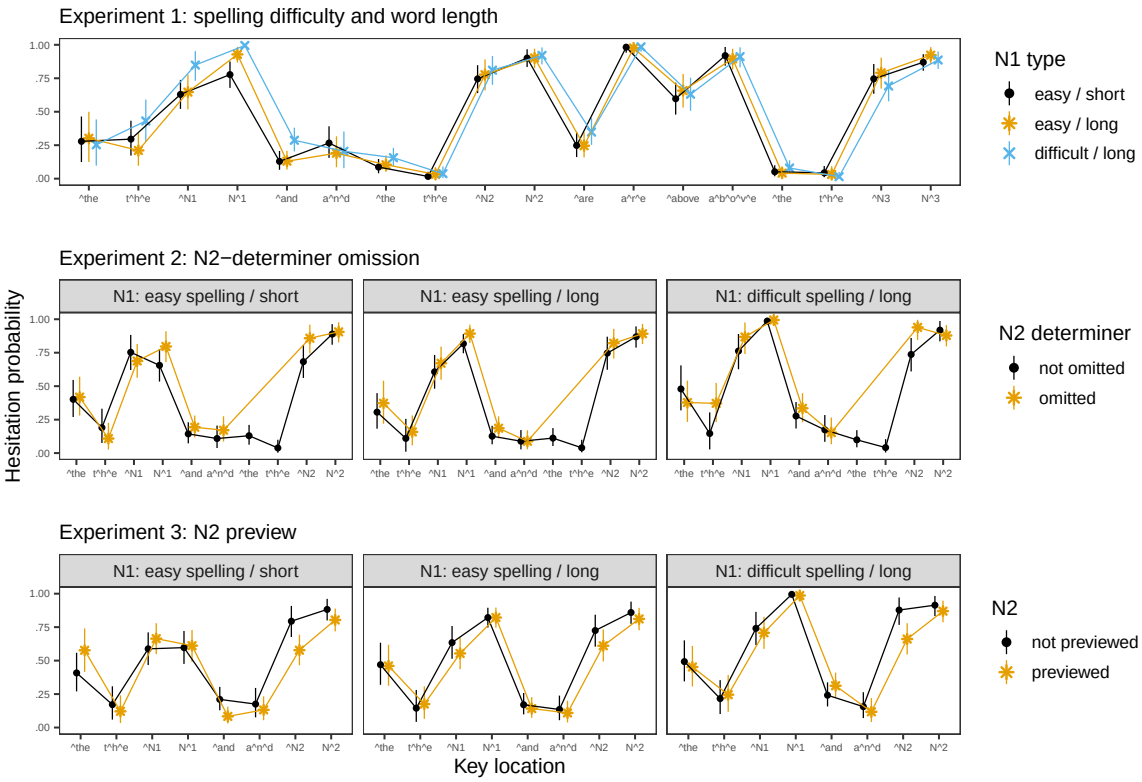

Figure 2: Hesitation probability. "^" indicate key location; "^the" indicates the transition from space to <t>; "t^h^e" are mid-word key-intervals. Errorbars are 95% PIs.

References

Roeser, J., Conijn, R., Chukharev, E., Ofstad, G. H., & Torrance, M. (2025). Typing in tandem: Language planning in multi-sentence text production is fundamentally parallel. *Journal of Experimental Psychology: General*, *154*(7), 1824–1854. https://doi.org/10.1037/xge0001759